JNTUH USED Reg 12-18 AM Set

Code No: 117EF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, November/December - 2018
MACHINE LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What can be the performance measure for checker learning program? [2]
b) What is a well-defined learning problem? [3]
c) State Minimum description length principle. [2]
d) Why decision trees over fit? [3]
e) Give example for Bayesian belief network. [2]
f) Differentiate between sample error and true error. [3]
g) What is meant by inverted deduction? [2]
h) Why case based reasoning is known as lazy learner? [3]
i) Differentiate between inductive learning and deductive learning. [2]
j) Give an example for mutation operator. [3]

PART-B

(50 Marks)

2. Explain candidate-elimination learning algorithm using version space. Will this algorithm converge to the correct hypothesis? Give illustrative examples. [10]

OR

3. Discuss in detail some successful applications of machine learning. [10]

4. Describe the properties of artificial neural nets. Why are they better suited for classification of objects/inputs? [10]

OR

5. Design a two-layer network of perceptrons that implements A XOR B. [10]

6. State Baye's theorem. Explain the two roles for Bayesian methods and discuss brute force MAP hypothesis learner. [10]

OR

7. What are the problems in estimating error? Explain the concept of confidence intervals for observed hypothesis error. [10]

8. Explain locally weighted regression methods as instance-based algorithm. [10]

OR

9. When to consider nearest neighbour learning? Discuss its advantages and disadvantages. [10]

10. How to use prior knowledge to reduce space complexity? Give illustrations. [10]

OR

11. Explain how genetic algorithms can be used for learning. Explain with an example. [10]

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